

# Numbers

Numbers are symbols that are used to represent a quantity of something. They have been used for thousands of years to answer the question “how many?”. At first people only used whole numbers (integers), but then came the idea of fractions and negative numbers.

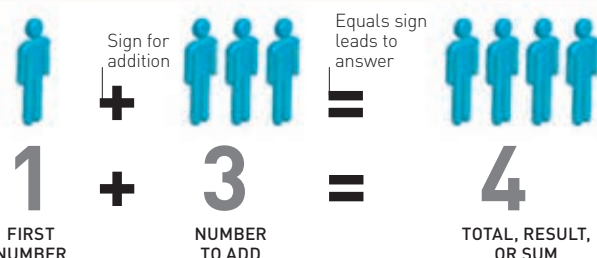
## NUMBER SYMBOLS

Many ancient civilizations used some form of number system. The modern Hindu-Arabic system is the simplest and most useful for mathematical calculations.

| Modern Hindu-Arabic | 1 | 2  | 3   | 4    | 5     | 6      | 7       | 8        | 9         | 10         |
|---------------------|---|----|-----|------|-------|--------|---------|----------|-----------|------------|
| Mayan               | • | •• | ••• | •••• | —     | —•     | —••     | —•••     | —••••     | ==         |
| Ancient Chinese     | 一 | 二  | 三   | 四    | 五     | 六      | 七       | 八        | 九         | 十          |
| Ancient Rome        | I | II | III | IV   | V     | VI     | VII     | VIII     | IX        | X          |
| Ancient Egypt       |   |    |     |      |       |        |         |          |           | ∩          |
| Babylonian          | 𐎶 | 𐎶𐎶 | 𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶 | 𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶𐎶 |

## ADDITION

Numbers can be added together to find the total of two or more quantities. Additions are written as equations by placing “+” between the numbers being added.



## SUBTRACTION

Subtraction is a mathematical way of working out how many are left if you take some of an original quantity away. It uses the “-” sign.

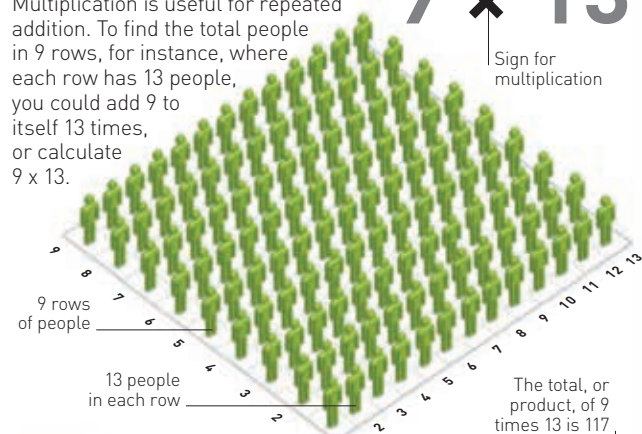


## MULTIPLICATION

Multiplication is useful for repeated addition. To find the total in 9 rows, for instance, where each row has 13 people, you could add 9 to itself 13 times, or calculate  $9 \times 13$ .

$$9 \times 13$$

Sign for multiplication



$$9 \times 13 = 13 + 13 + 13 + 13 + 13 + 13 + 13 + 13 + 13 = 117$$

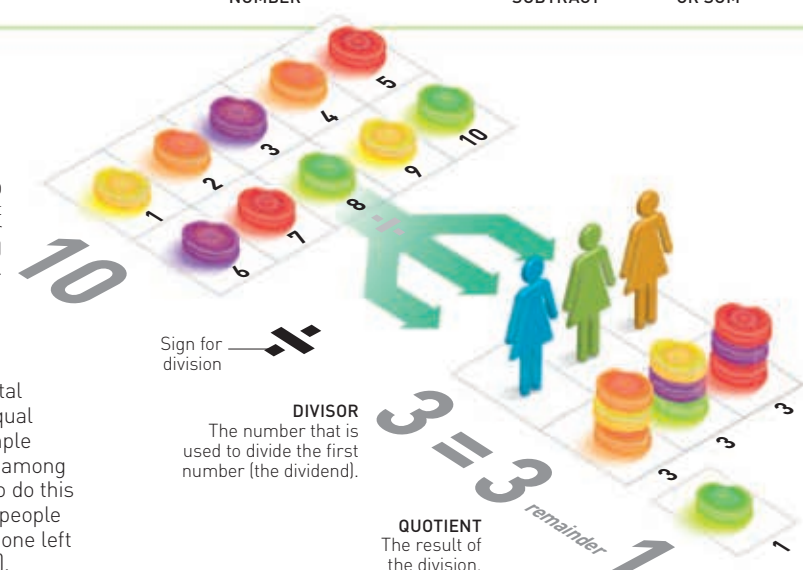
$$10 \div 3$$

Division sign

**DIVIDEND**  
This is the number that is being divided by (or shared out among) another number.

## DIVISION

Division is used to divide up a total number of things into several equal bundles, or amounts. This example shows how to divide ten sweets among three people. It is not possible to do this evenly, so after giving the three people three sweets each, there is still one left over (known as the “remainder”).

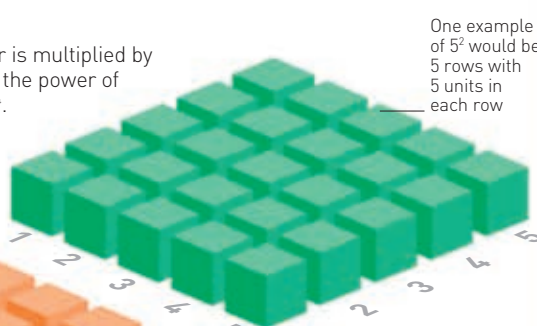


## POWERS

A “power” is the number of times a number is multiplied by itself. So “ $5 \times 5 \times 5 \times 5$ ” is said to be “five to the power of four” which is written mathematically as  $5^4$ .

**5<sup>3</sup>** This is the power, which shows how many times to multiply the number ( $5^3$  means  $5 \times 5 \times 5$ )

This is the number that the power relates to



SQUARED NUMBER

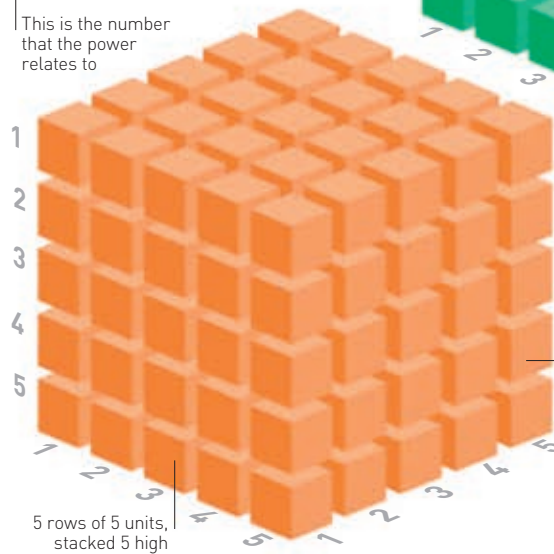
$$5 \times 5 = 5^2 = 25$$

$5^2$  is called “5 squared”

CUBED NUMBER

$$5 \times 5 \times 5 = 5^3 = 125$$

$5^3$  is called “5 cubed”

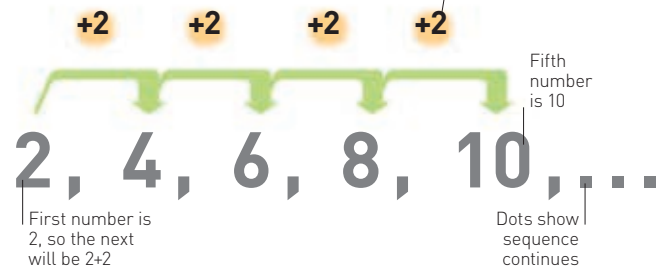


## NUMBER SEQUENCES

A sequence of numbers is a series of numbers that follow one another according to a pattern, such as each number being two higher than the previous term.

Each number in this sequence is 2 higher than the number before it

**A BASIC SEQUENCE**  
The rule for this sequence is that each number equals the previous number plus 2.



**FIBONACCI SEQUENCE**  
This is a very famous number sequence that appears in lots of natural formations such as flower petals and spiral galaxies.

Each number in this sequence is the sum of the two numbers before it

